COMP 3920 Database Systems 1

MySQL
Import the Data

Introduction:
The goal of this document is to serve as a walkthrough importing data into MySQL 8.

For this document, we will be using command prompt commands and MySQL Workbench in MySQL 8.

We are going to use 3 ways to import data into MySQL:

1. Run mysql from the command line and provide an SQL file for input.
2. Log in to mysql and use the source command to provide an SQL file for import.
3. Open an SQL file in MySQL Workbench's Run Script Tool and execute.

All 3 methods will be using the Employees Database provided by MySQL on their website:
https://github.com/datacharmer/test_db
Instructions and details available here:
https://dev.mysql.com/doc/employee/en/employees-introduction.html

Step 1:
Download `EmployeeDB.zip` and unzip.

Step 2 - Import using MySQL from command prompt (Method 1):
Open the Command Prompt and change directory to the folder that contains the employee database you extracted from the previous step.

We are going to use the following command:
```
mysql -u root -p -t < EmployeesDB.sql
```
Notice a few new things.
There is a `-t` option. All this does is use the 'table' format to make the output a bit prettier.
The next thing that's different is the `< EmployeesDB.sql`.
Normally when we run mysql from the command line, MySQL comes back with its own command prompt waiting for SQL statements from the keyboard. In this case we want the SQL statements to come from a file instead of the keyboard, so we use the Windows Command Prompt to *redirect* the input from the keyboard to use a file instead. This is called redirection.

Submit your screenshot:
Filename: **03_Method_1_Import.jpg**

Submit

```
C:\Users\User\Downloads\test_db-master>mysql -u root -p -t < employees.sql
Enter password: ****************
+----------------------------+
| INFO                       |
+----------------------------+
| CREATING DATABASE STRUCTURE |
+----------------------------+
```

You will see output as each table is created and data is imported into those tables. This may take a while (2-3 minutes).

You should see this when it's complete.

```
+------------------+
| INFO             |
+------------------+
| LOADING salaries |
+------------------+
+---------------------+
| data_load_time_diff |
+---------------------+
| 00:02:25            |
+---------------------+

C:\Users\User\Downloads\test_db-master>
```

We have completed Method 1 to import the data.
Let's verify that the data is there.

```
mysql -u root -p
SHOW DATABASES;
SHOW TABLES FROM employees;
```

Submit your screenshot:

Filename: **04_Method_1_Show_Tables.jpg**

```
C:\Users\User\Downloads\test_db-master>mysql -u root -p
Enter password: *****************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 47
Server version: 8.0.11 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
[redacted]
8 rows in set (0.10 sec)

mysql> show tables from employees;
[redacted]
8 rows in set (0.04 sec)

mysql> _
```

Step 3 - Import using `source` command (Method 2):
We will be re-importing the same data again so we'll remove the database we created in the previous step.
Log in to mysql:
```
mysql -u root -p
```

Drop the Database:
```
DROP DATABASE employees;
```

You can confirm the database is gone by running:
```
SHOW TABLES FROM employees;
```

```
mysql> drop database employees;
Query OK, 8 rows affected (2.44 sec)

mysql> show tables from employees;
ERROR 1049 (42000): Unknown database 'employees'
mysql>
```

Now for Method 2.
To use the `source` command simply type `source filename.sql` (while logged in to MySQL).
Like so:
```
source EmployeesDB.sql
```
The output will be similar to Method 1, except the source command is more verbose and will give you updates for each set of rows being updated.

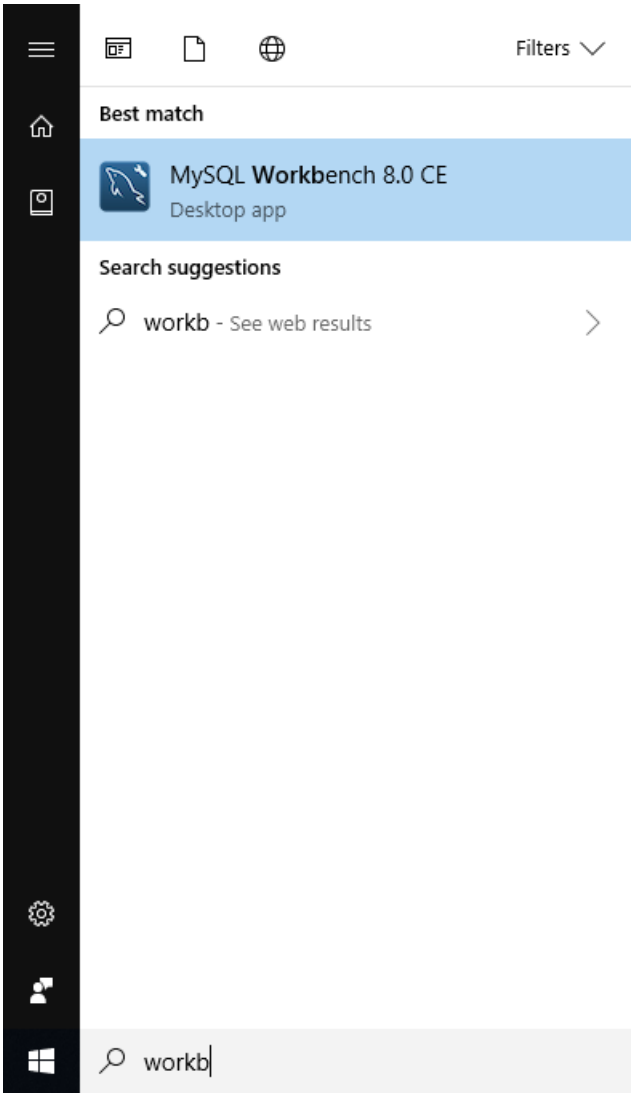
Submit your screenshot:
Filename: **05_Method_2_Import.jpg**

```
Query OK, 7671 rows affected (0.33 sec)
Records: 7671  Duplicates: 0  Warnings: 0

+---------------------+
| data_load_time_diff |
+---------------------+
| 00:02:20            |
+---------------------+
1 row in set (0.14 sec)

mysql>
```

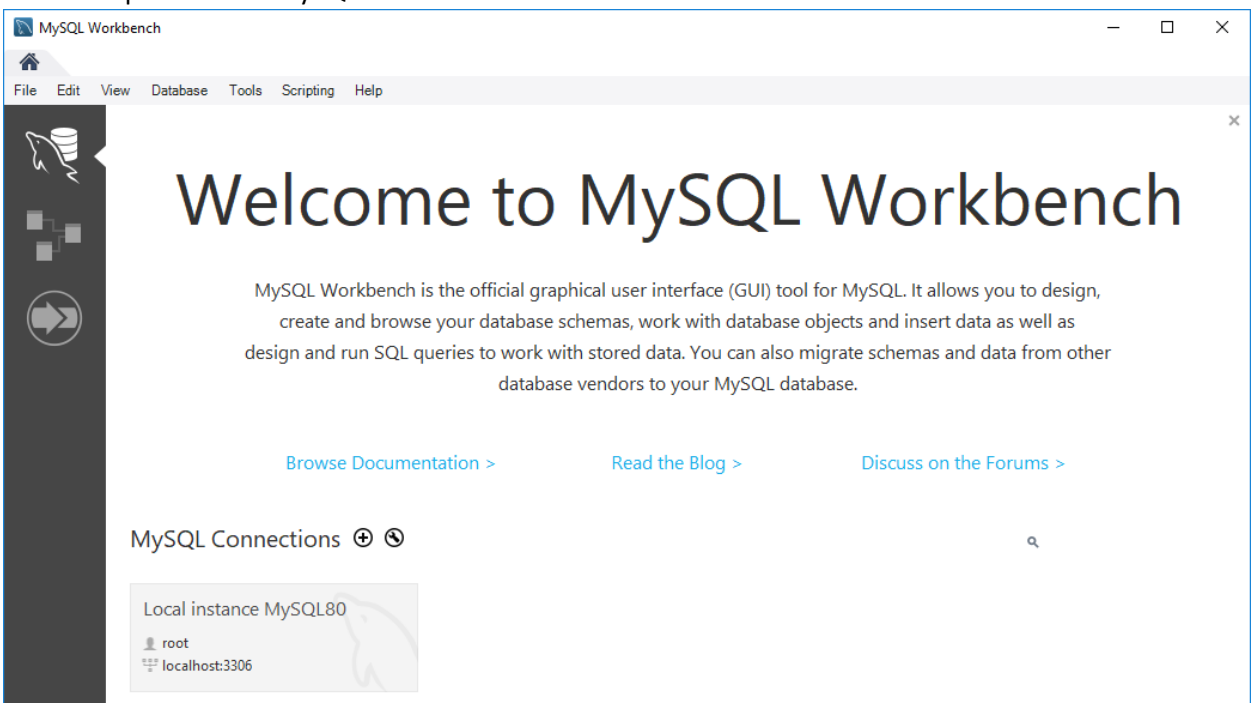
Step 4 – Using MySQL Workbench (Method 3):
Before opening MySQL Workbench, let's delete the database created in the previous step so we can re-import it.

```
DROP DATABASE employees;
```

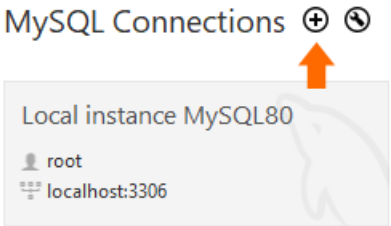
Open MySQL Workbench
Typing `workb` in the start menu was enough to find it for me.



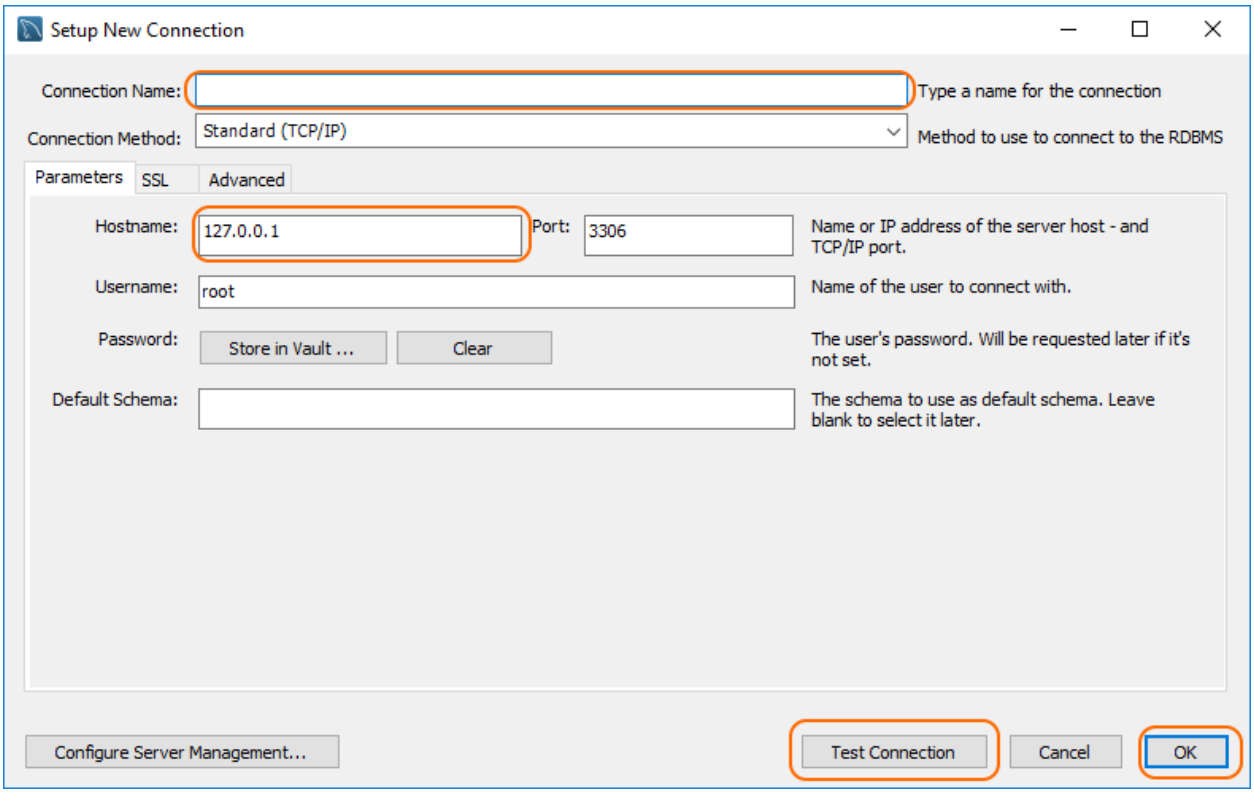
The startup screen for MySQL Workbench looks like this:



If you don't have a “Local instance MySQL80” connection button on the bottom right create a new connection by clicking the + next to MySQL Connections.

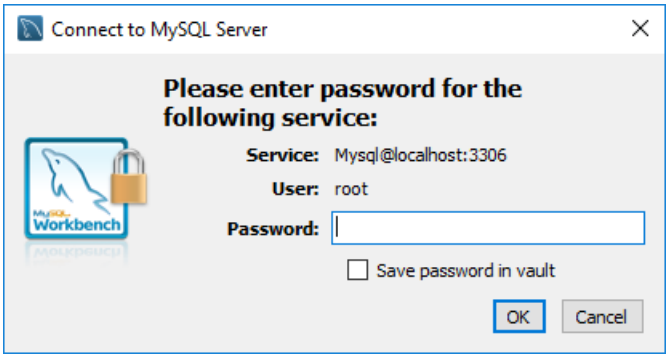


Creating a “Local instance MySQL80” connection button:
Give your new connection a name: Local instance MySQL80
Leave the Hostname as 127.0.0.1 (or you can use localhost – same thing).
Keep the default port 3306.
Click Test Connect to see if it works (it will prompt you for a password).
Click Ok.

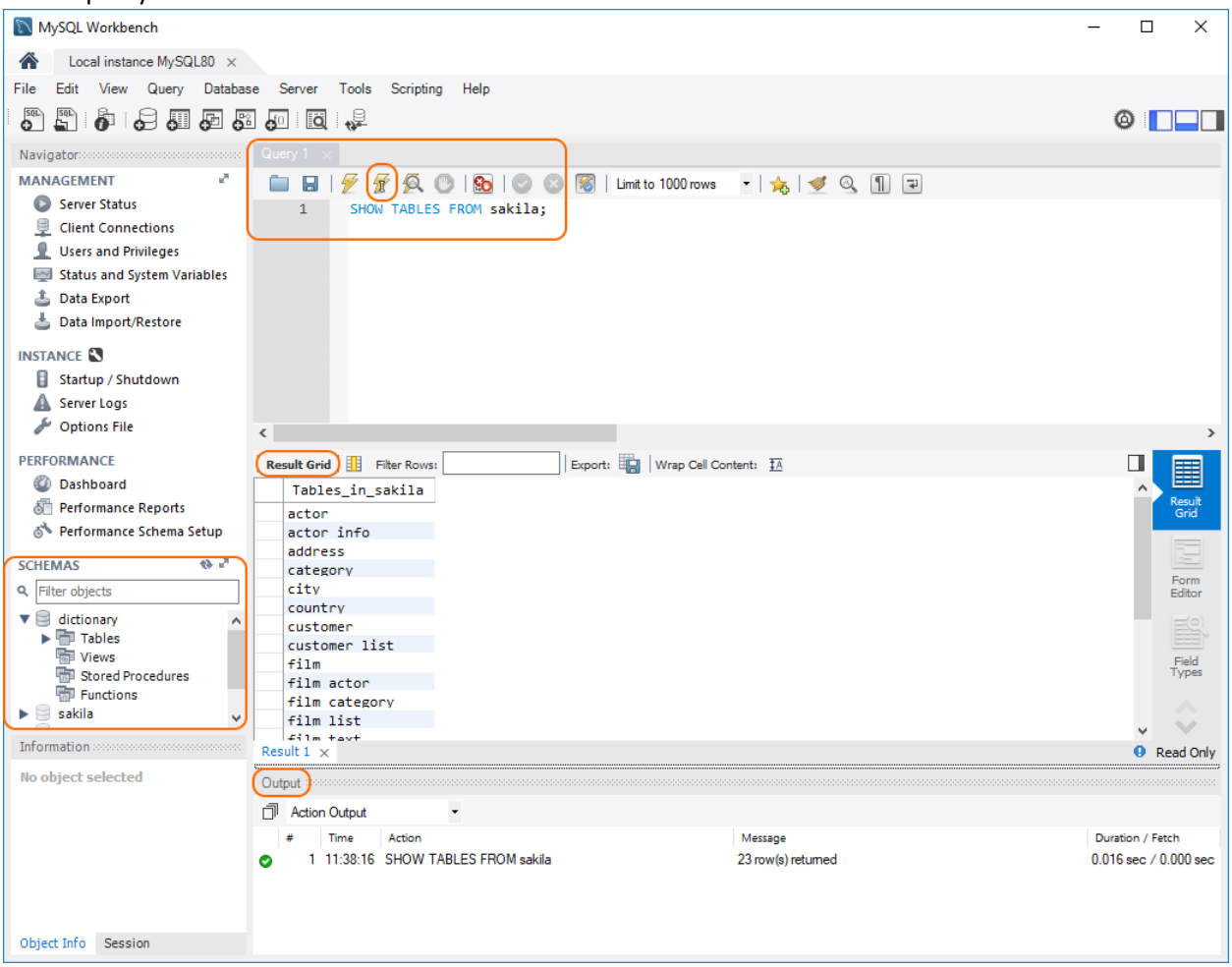


Open up Local instance MySQL80.

Double click the “Local instance MySQL80” connection button and enter your root password when prompted.



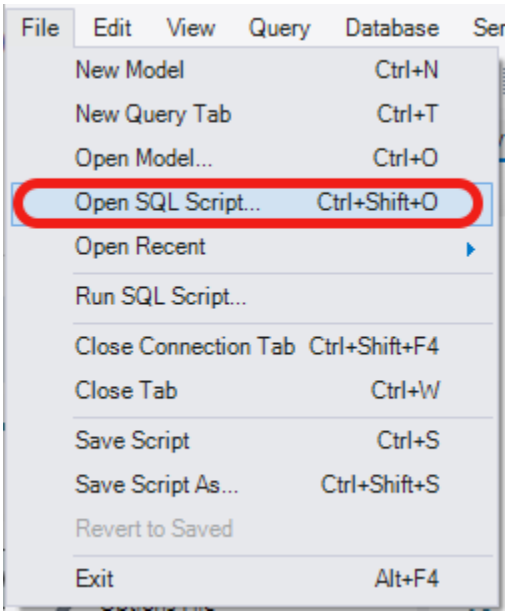
Once open you should see a screen like this:



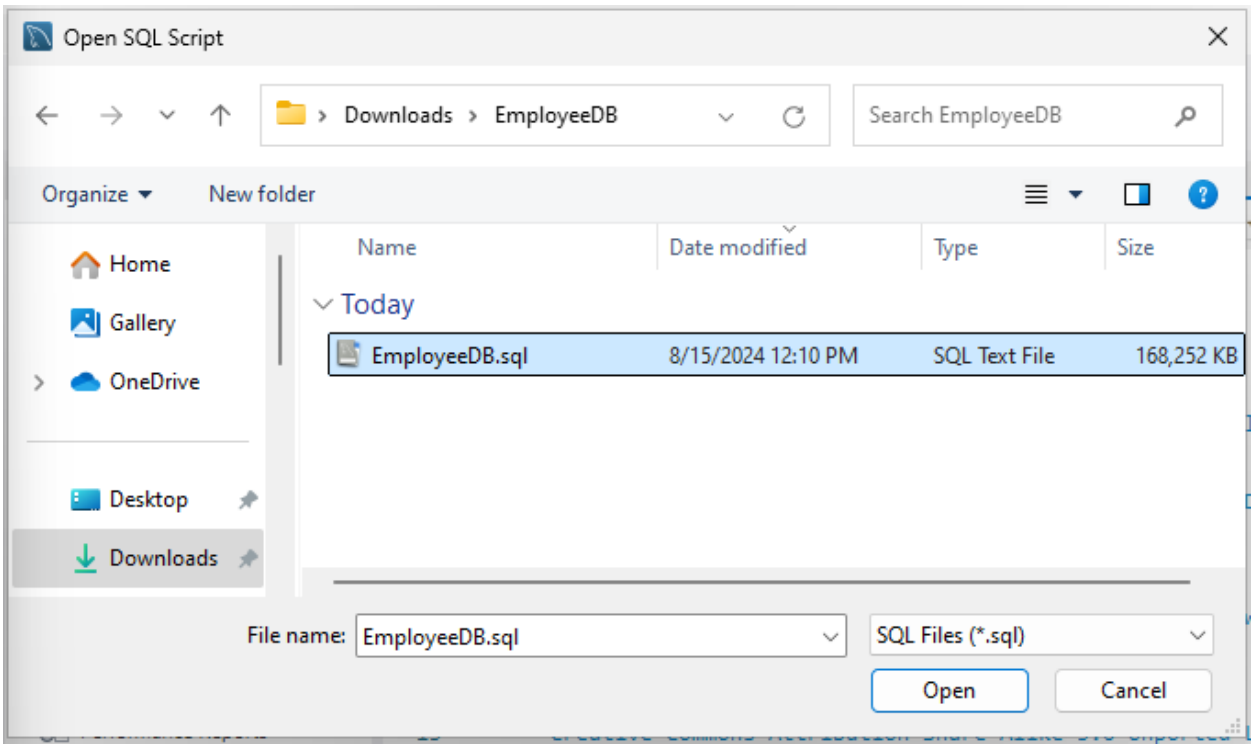
At this point I'll point out 4 main parts of the screen.

1. There is a Schema (or Database) Section on the bottom right that shows all the databases, tables, views, procedures and functions with the databases.
2. At the top there is a Query Section for typing in your queries or opening an SQL file. You can click on the lightning bolt with the cursor to execute a single query at a time.
3. Once you execute a query (for example a SELECT or SHOW TABLES) you'll see the results in the Result Grid below the Query Section
4. The Output Section shows additional information about the queries that have run. It will show the number of rows returned, how long the query took to execute or it will show an error and a hint to find your syntax error in your query.

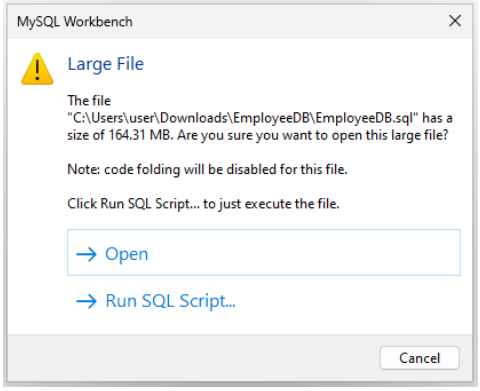
For our import we will open our `EmployeeDB.sql` script by using the File → Open SQL Script… from the menu.



Find `EmployeeDB.sql` and click Open.

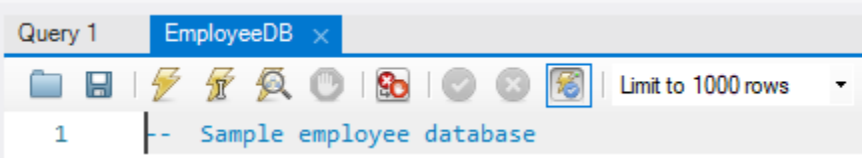


You may get a warning that the file is too large and that it would be better to run this file using "Run SQL Script". I have had bad experiences with "Run SQL Script" as it often crashes on Windows and Linux. There are several bugs in the "Run SQL Script" functionality that MySQL Workbench hasn't fixed yet.



Click "Open".

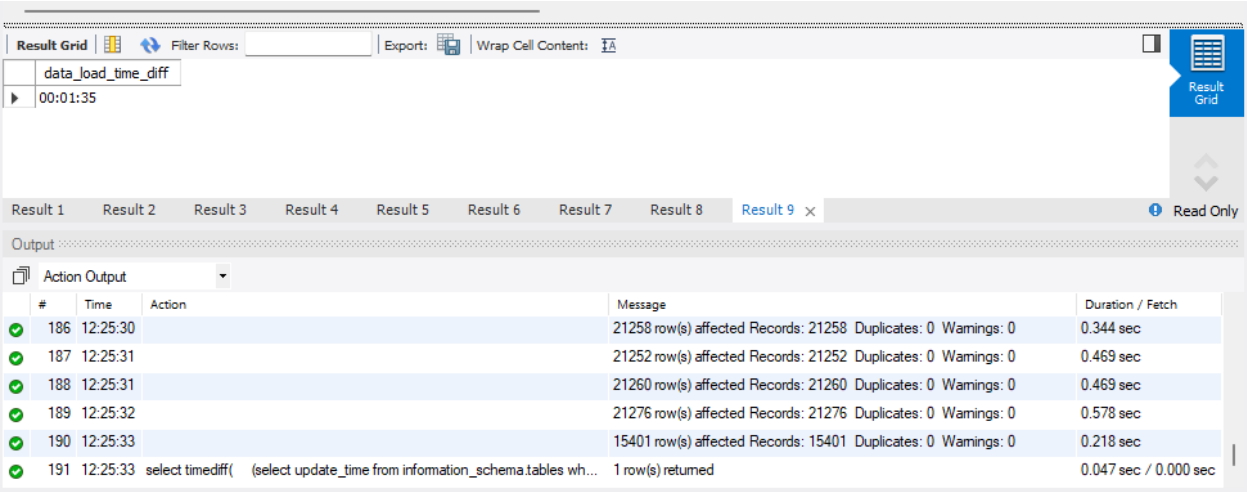
Click on the lightning bolt icon to execute the file.



This might take a while, so be patient - this is a large file!

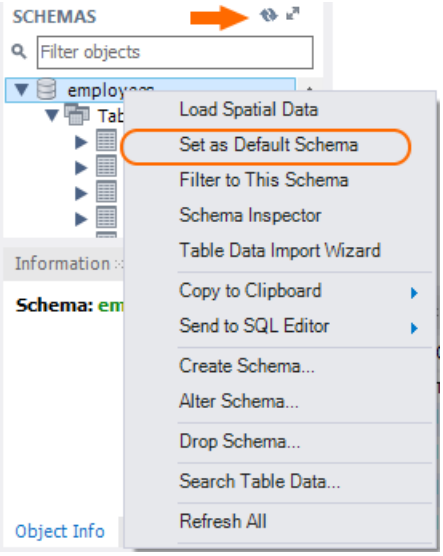
Workbench may not give you much feedback (if at all) while it is running in the background.
You'll have to wait roughly the same amount of time as the other import methods.

Once finished you should see an output section that looks like this:

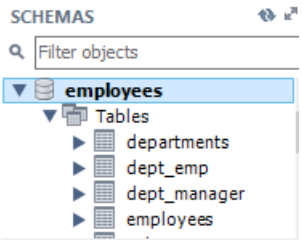


To confirm that the database loaded correctly we are going to go to the Schema Section:
Click Refresh if you don't see the `employees` database below.

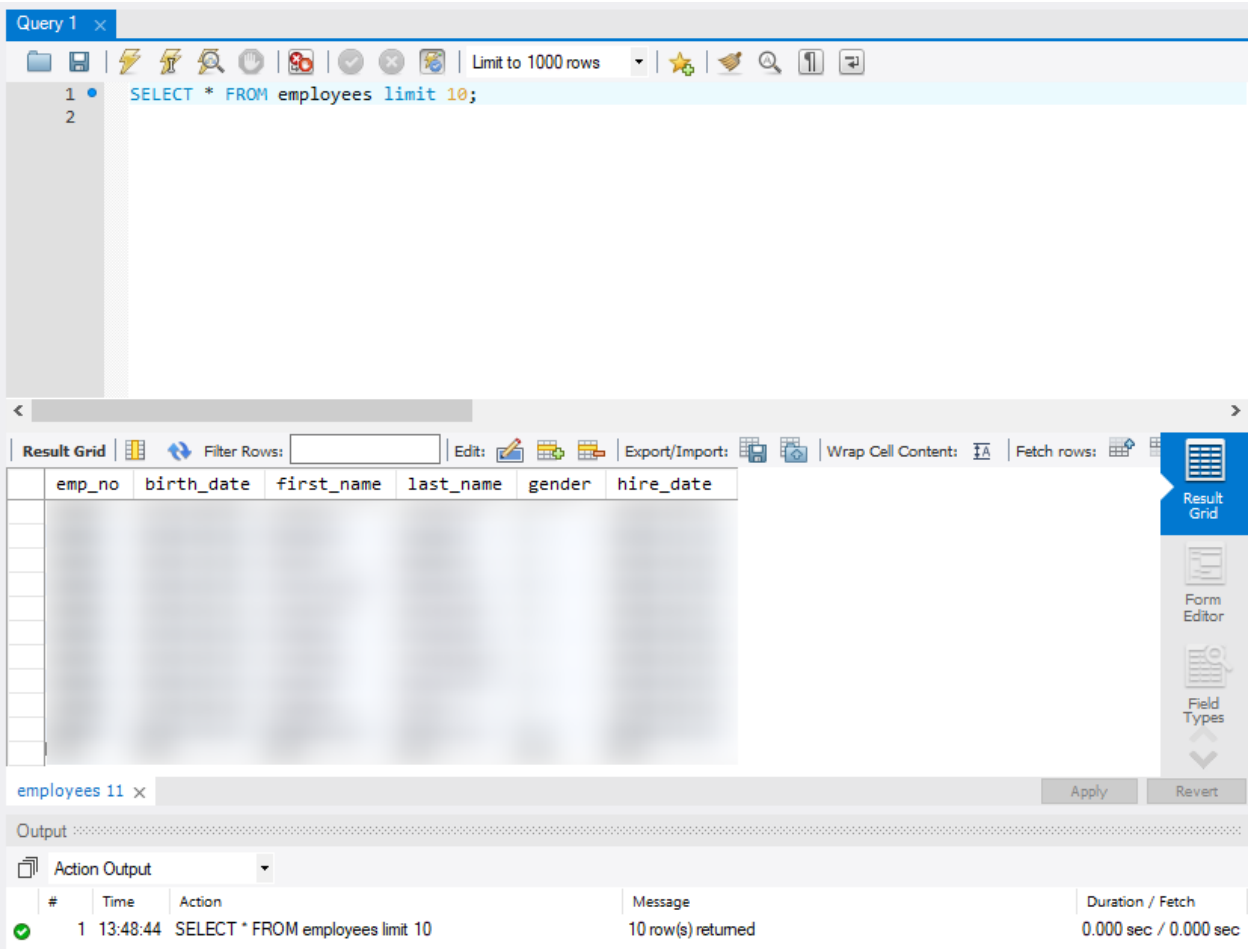
Then set `employees` as your Default Schema:



You should see the `employees` database shown in **bold**.



Let's look at the data.
In the Query Section execute the following query:

```
SELECT * FROM employees LIMIT 10;
```

Submit your screenshot:
Filename: **06_Method_3_Workbench.jpg**